KP Smart Crop Doctor: An Explainable, Knowledge-Grounded, Bilingual Plant-Disease Decision-Support System for Khyber Pakhtunkhwa

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***Abstract*—Foliar diseases threaten smallholder yields in Khyber Pakhtunkhwa (KP), Pakistan, where extension officers are scarce and diagnosis is time-critical. Existing deep-learning diagnostics return opaque labels, while generic large language models can hallucinate unverified chemical advice, and neither serves Urdu-speaking farmers well. We present *KP Smart Crop Doctor*, a CPU-deployable decision-support prototype for 11 disease and healthy classes of apple, maize and potato. A MobileNetV3-Large classifier trained on a leaf-group-isolated split (N=9,175) reaches 97.53% Top-1 accuracy, 99.78% Top-2 accuracy, 0.9657 macro-F1 and 1.40% expected calibration error (ECE) on a held-out test set, outperforming an EfficientNet-B0 baseline. Grad-CAM on the final convolutional layer gives spatial explanations, checked with a parameter-randomisation test. The predicted crop and disease condition a FAISS retrieval over 11 KP extension bulletins (Hit@1 96.97%, Hit@3 100%, MRR 0.9848), and a deterministic extractive synthesiser produces cited advisories in English and Urdu (RTL). Out-of-scope crops are rejected in 4/4 cases. The full pipeline runs in 184.22 ms on average on a CPU. The system is an advisory prototype, not a replacement for extension officers, and has not been validated on field imagery.**

***Index Terms*—plant disease classification, transfer learning, Grad-CAM, explainable AI, retrieval-augmented generation, agricultural decision support, Urdu, Khyber Pakhtunkhwa**

I. Introduction

Apple, maize and potato underpin rural livelihoods across KP: apples in Swat and Hazara, maize on the central plains of Mardan and Swabi, and potato in the highland valleys. All three are exposed to fungal, bacterial and oomycete leaf diseases whose early visual symptoms overlap, so correct identification depends on trained judgement that farmers often cannot reach within the window in which a treatment decision matters.

Image-based deep learning can recognise crop diseases accurately under laboratory conditions [1], but a bare class label is neither actionable nor auditable. A generative language model could add advice, but ungrounded generation is a known failure mode [2], and an invented pesticide or dosage carries real cost. Finally, advice delivered only in English excludes many intended users.

This paper describes a system that addresses these three gaps jointly. Its contributions are:

1) A leak-controlled 11-class KP crop-disease benchmark (N=9,175) built with leaf-group isolation and perceptual hashing, and a baseline comparison of two edge-oriented backbones with calibration analysis.
2) Grad-CAM explanations on a lightweight backbone, with a cascading parameter-randomisation sanity check and its limitations reported.
3) A provenance-tracked retrieval layer over 11 KP extension bulletins, with a crop-scope guardrail and a deterministic extractive synthesiser that cannot introduce unretrieved dosages or trade names.
4) An entity-preserving English–Urdu localisation layer with right-to-left rendering, integrated with the above in a Streamlit application and profiled end-to-end on CPU.

II. Related Work

Plant-disease recognition. Mohanty *et al.* showed that CNNs trained on the PlantVillage corpus [3] classify leaf diseases with high accuracy, while noting a sharp drop on images captured outside the training conditions [1]. We keep that caveat central: our accuracy is laboratory-condition accuracy. Efficient backbones such as MobileNetV3 [4] and EfficientNet [5] are the usual choice when inference must run on modest hardware.

Explainability. Grad-CAM [6] localises class evidence using gradients at a late convolutional layer without architectural change. Adebayo *et al.* showed that some saliency maps are insensitive to the model's parameters and proposed randomisation tests to detect this [7]; we apply such a test.

Retrieval-augmented generation. RAG conditions generation on retrieved passages from an inspectable external memory [8]. We use dense sentence embeddings [9] with an exact inner-product FAISS index [10], but replace the generator with extractive synthesis so that every sentence shown to the user is traceable to a source passage. To our knowledge, prior work rarely combines classification, explanation, provenance-linked retrieval and Urdu output for KP crops in a single evaluated pipeline.

III. System Architecture

Fig. 1 shows the decoupled pipeline. A leaf image is resized to 256×256, centre-cropped to 224×224 and ImageNet-normalised. MobileNetV3-Large returns Top-1 and Top-2 diagnoses with confidences; PyTorch hooks capture activations and gradients at `features.16` to produce a Grad-CAM overlay. The predicted crop and disease form a retrieval query

TABLE I
CLASS TAXONOMY AND SPLIT DISTRIBUTION (N=9,175)

Crop	Class	Train	Val	Test
Apple	Scab	441	94	95
	Black rot	435	93	93
	Cedar apple rust	192	41	42
	Healthy	1,152	247	246
Maize	Gray leaf spot	359	77	77
	Common rust	834	179	179
	Northern leaf blight	688	150	147
	Healthy	813	174	175
Potato	Early blight	700	150	150
	Late blight	700	150	150
	Healthy	108	21	23
Total		6,422	1,376	1,377

that passes a scope guardrail (crop whitelist plus a cosine-similarity threshold τ=0.45) before searching the FAISS index. Retrieved passages are assembled into a structured advisory (symptoms, likely cause, integrated management, cultural prevention, crop-stage guidance) with citations, localised into Urdu, and displayed in a Streamlit dashboard with an opacity slider, tabbed cards and a follow-up question box.

IV. DATASET AND LEAKAGE CONTROL

The corpus contains 9,175 foliar images over 11 classes (Table I); the label taxonomy follows the PlantVillage convention [3]. Multiple photographs of one physical leaf are a common source of inflated test accuracy, so three safeguards were applied before any training: (i) images sharing a leaf session were grouped by file-naming pattern; (ii) 64-bit difference hashes (dHash) were computed and image pairs with Hamming distance ≤ 4 were forced into the same partition; and (iii) a group-stratified 70/15/15 split (GroupKFold on `leaf_id`) was used. Automated tests found no leaf cluster shared between train, validation and test. This rules out overlap detectable by file grouping and the dHash threshold; it cannot exclude near-duplicates that both checks miss.

V. VISION MODEL AND EVALUATION

A. Training protocol

MobileNetV3-Large [4] and EfficientNet-B0 [5] were fine-tuned from ImageNet weights under identical conditions: seed 42, AdamW [11], cosine-annealed learning rate [12], class-weighted cross-entropy to counter imbalance (e.g. 108 healthy-potato training images), and up to 10 epochs (Fig. 2). The model was selected on validation macro-F1 *before* the test set was evaluated.

B. Results

Table II shows that MobileNetV3-Large is more accurate, faster and better calibrated than the baseline, while being only marginally larger on disk. On 1,377 held-out images it makes 34 errors (2.47%). Calibration follows Guo *et al.*'s ECE [13]: the reliability curves in Fig. 3(b) lie above the diagonal, i.e. confidence is slightly conservative rather than overconfident on in-distribution data.

TABLE II
BACKBONE COMPARISON ON THE HELD-OUT TEST SET (N=1,377)

Metric	MobileNetV3-L	EfficientNet-B0
Validation macro-F1	**0.9773**	0.9535
Validation accuracy	**98.18%**	96.08%
Test Top-1 accuracy	**97.53%** (1,343)	95.06% (1,309)
Test Top-2 accuracy	**99.78%** (1,374)	99.35% (1,368)
Test macro-F1	**0.9657**	0.9352
Parameters / size	4.22 M / 16.28 MB	4.02 M / 15.63 MB
CPU latency per image	**28.38 ms** (35.2 FPS)	48.01 ms (20.8 FPS)
ECE	**0.0140**	0.0589

TABLE III
GRAD-CAM PARAMETER-RANDOMISATION TEST (TRAINED VS. RANDOMISED)

Class	Pearson r	Spearman ρ	MAE	Verdict
Maize gray leaf spot	0.0000	−0.0410	0.4097	Pass
Potato early blight	0.0000	−0.0084	0.2952	Pass
Apple scab	0.0000	0.1208	0.2381	Pass
Potato healthy	0.0000	0.1859	0.2539	Pass

C. Error analysis

All 34 errors stay within a crop; there is no cross-crop confusion. Errors fall into four groups: maize gray leaf spot vs. northern leaf blight (23 cases, 67.6%), whose early lesions are both elongated grey-tan necrosis (gray-leaf-spot recall is 0.81, Fig. 3(a)); healthy apple leaves with specular glare or dust read as scab (4, 11.8%); incipient potato late blight covering under 2% of the leaf and predicted healthy (3, 8.8%); and other intra-crop confusions such as atypical scab resembling black rot (4, 11.8%). Because the dominant confusion is between two visually similar maize diseases, the interface always shows the Top-2 alternative.

VI. EXPLAINABILITY

Grad-CAM [6] is applied at `features.16`, the last convolution (7×7×960); activations are weighted by globally pooled class gradients, passed through ReLU, upsampled and overlaid. Fig. 4 shows the overlays for all 11 classes. For disease classes the salient regions generally coincide with lesions (e.g. potato blight, maize gray leaf spot); for healthy leaves salience is diffuse or on margins, which is itself informative to a reviewer.

Sanity check. Following Adebayo *et al.* [7], we recomputed Grad-CAM on a model with randomised parameters and compared the maps (Table III, Fig. 5). The explanations changed completely, which indicates that they depend on learned weights. A caveat: the randomised model's maps are constant, so Pearson r is degenerate (reported as 0.0000) and the Spearman ρ and MAE columns carry the comparison. The test was run on four images; a larger sample would strengthen the claim.

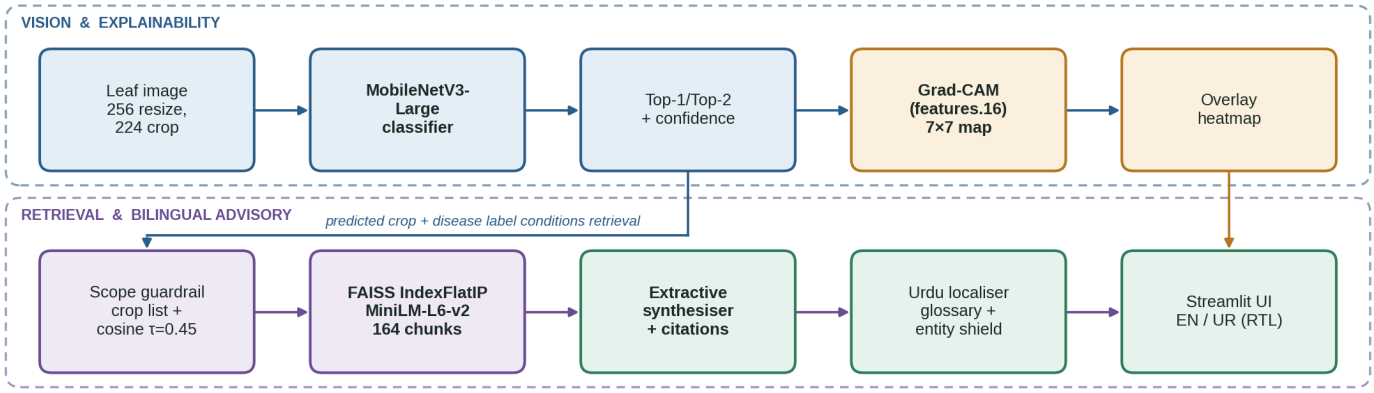


Fig. 1. System architecture of KP Smart Crop Doctor. The vision branch (top) conditions the retrieval branch (bottom); the heatmap is shown beside the advisory.

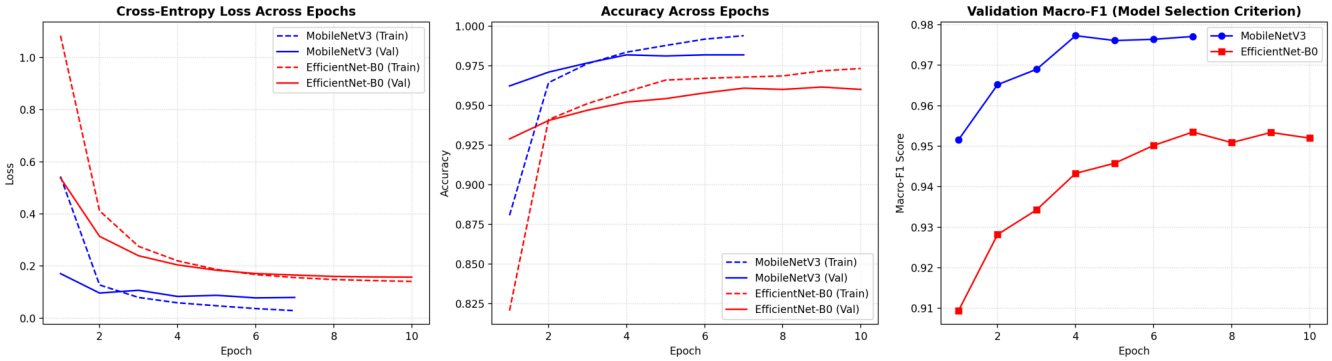


Fig. 2. Training and validation behaviour of the two backbones. MobileNetV3 attains the higher validation macro-F1 (selection criterion) throughout.

TABLE IV
RETRIEVAL OVER 33 DOMAIN QUERIES

Metric	Result	Definition
Hit@1	96.97% (32/33)	target bulletin at rank 1
Hit@3 / Hit@5	100% (33/33)	within top 3 / 5 chunks
MRR	0.9848	mean reciprocal rank
Precision@1 / @3	96.97% / 79.80%	relevant chunks in top k
Retrieval latency	21.84 ms	embedding + search, CPU

VII. Knowledge Base, Retrieval and Bilingual Advisory

A. Knowledge base and retrieval

The knowledge base comprises 11 extension bulletins from ARI Tarnab, CCRI Pirsabak, PRC Abbottabad and the KP Department of Agriculture Extension [14], one per disease class plus crop-level practice guides. Text was split into 164 chunks (at most 500 characters, 50 overlap) and embedded with all-MiniLM-L6-v2 into 384-d vectors [9], indexed with exact inner-product FAISS [10]. Every chunk keeps its bulletin identifier (`KP-AGRI-*`) as provenance. On 33 domain queries retrieval reached the results in Table IV.

B. Grounded synthesis and guardrails

To rule out generative chemical hallucination, advisories are built by a deterministic extractive synthesiser that copies structured sections from retrieved passages and attaches the source citation; no text is generated freely. Groundedness, citation correctness and evidence faithfulness were all 100% on the 15-case advisory benchmark. Because extraction is faithful by construction, this figure verifies the implementation rather than measuring a generative model's behaviour; the real constraint is coverage, i.e. what the 11 bulletins contain. A guardrail rejects queries about unsupported crops (cotton, wheat, sugarcane, citrus), with 4/4 correct abstentions.

C. Urdu localisation

Advisories are localised with a 33-term agricultural glossary. Latin binomials (e.g. *Phytophthora infestans*) and citation keys are replaced by protection tokens before translation and restored verbatim afterwards, so pathogen and chemical names are never altered. The interface injects right-to-left CSS and Noto Nastaliq Urdu typography. Localisation changes text length by about 3%.

VIII. End-to-End Evaluation

The complete pipeline was run on 17 cases: 11 canonical, 2 difficult borderline and 4 out-of-scope. All 17 executed without unhandled exceptions. Diagnostic accuracy was 92.31% (12/13; all 11 canonical and 1 of 2 difficult cases), guardrail abstention 100% (4/4) and citation linkage 100% (13/13). Mean CPU latency from image to Urdu advisory

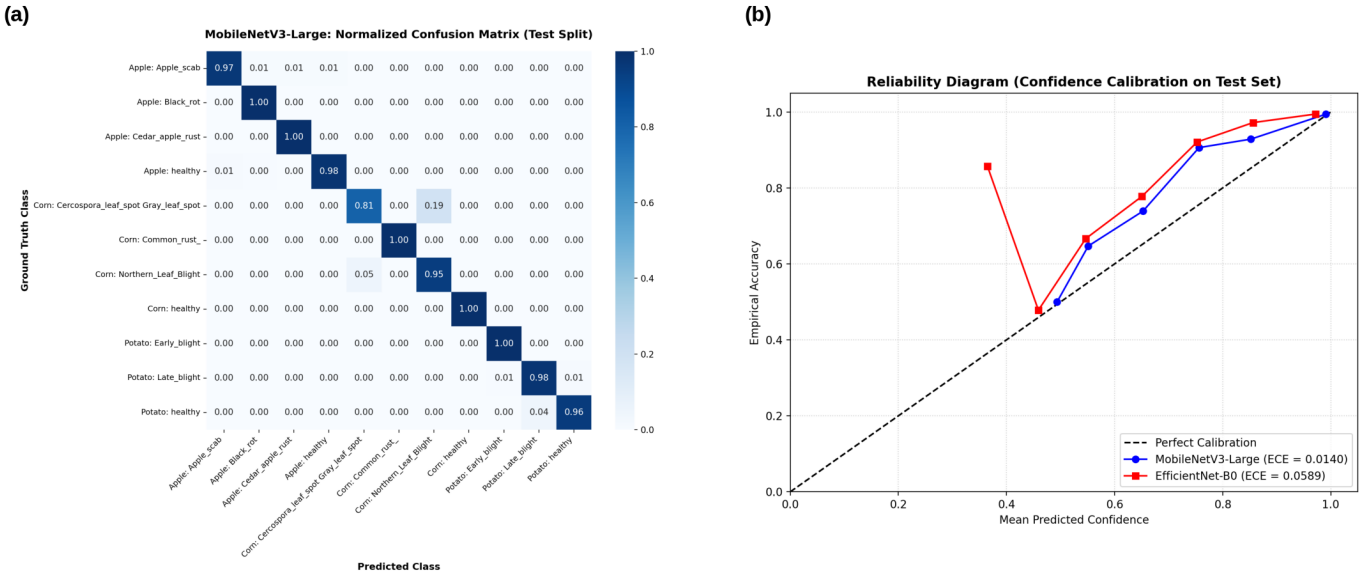


Fig. 3. (a) Row-normalised confusion matrix of MobileNetV3-Large on the test set. (b) Reliability diagram: MobileNetV3-Large (ECE 0.0140) vs. EfficientNet-B0 (ECE 0.0589).

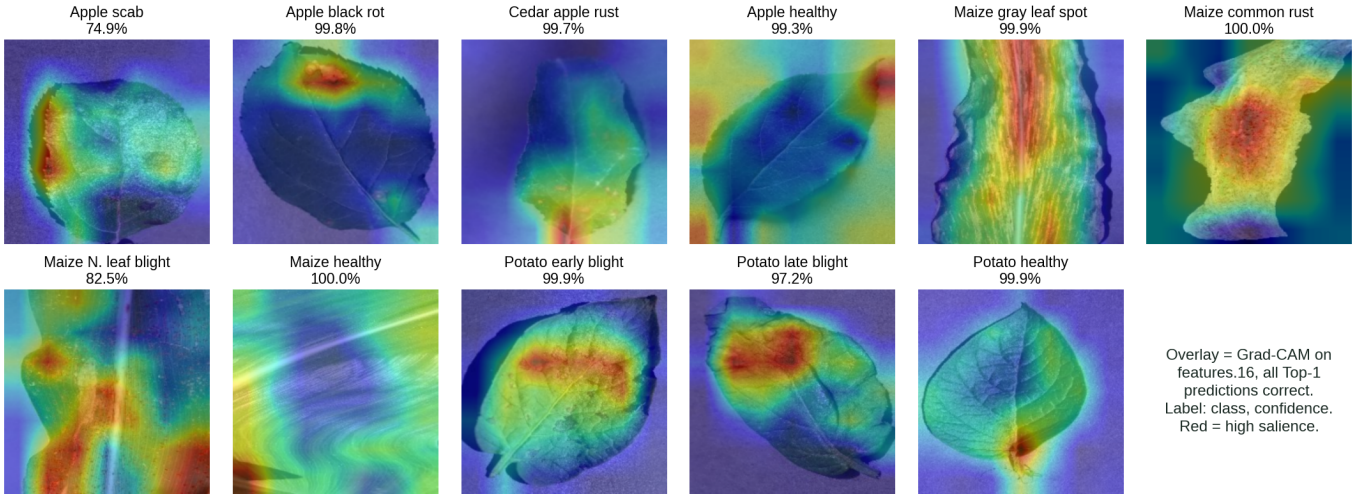


Fig. 4. Grad-CAM overlays on one correctly classified test image per class, with Top-1 confidence.

TABLE V
END-TO-END CPU LATENCY PROFILE (N=17)

Stage	Mean (ms)	P95 (ms)	Share
Vision inference (MobileNetV3)	71.56	116.52	38.8%
Grad-CAM generation	58.55	95.33	31.8%
FAISS retrieval (top-3)	15.94	18.88	8.7%
Advisory synthesis (EN)	18.62	22.09	10.1%
Urdu localisation + RTL	19.54	23.73	10.6%
Total	184.22	271.51	100%

was 184.22 ms (P95 271.51 ms), dominated by the two vision stages (Table V).

Ablations. (i) *Grad-CAM*: adding explanation costs about 59 ms (+270% over classification alone), acceptable for visual validation. (ii) *Metadata-filtered retrieval*: unfiltered semantic search leaked across crops in 13.33% of queries (e.g. potato early blight returned for a generic maize query); conditioning on the vision output reduced leakage to 0%. (iii) *Language*: English 20.77 ms vs. cached Urdu 19.54 ms. (iv) *Scope*: in-scope queries had mean similarity 0.5458 against τ=0.45; all out-of-scope queries were abstained via lexical and cosine filters. Latencies in Table V come from the end-to-end harness and differ from the per-image benchmark in Table II; absolute values should not be compared across the two.

IX. LIMITATIONS

(1) *Domain gap.* Training images are controlled laboratory captures; smartphone photographs with glare, weeds, motion blur or co-occurring infections may perform worse, and field readiness cannot be claimed without prospective trials in KP. Reported accuracy is an upper bound for field use. (2) *Closed*

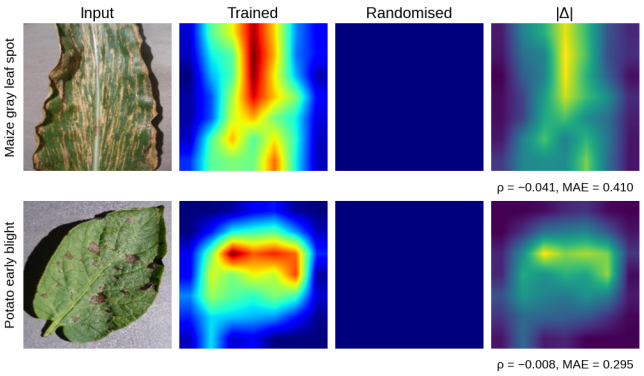


Fig. 5. Sanity check: trained-model map, randomised-model map and absolute difference for two classes.

set. Only three crops and 11 classes are supported; an unseen crop or a nutrient deficiency will be forced into the nearest class, and softmax may be overconfident on non-leaf inputs. The guardrail and Top-2 display mitigate but do not remove this. (3) *Explanation granularity.* A 7×7 map gives coarse regional salience and does not establish biological causality. (4) *Evaluation scale.* The end-to-end set (17 cases), advisory set (15) and sanity test (4 images) are small. (5) *Knowledge coverage.* Advice is limited to 11 bulletins; chemical products carry toxicological and environmental risk, so the tool supplements, and never replaces, licensed on-site assessment.

X. Conclusion and Future Work

KP Smart Crop Doctor shows that an edge-efficient classifier, spatial explanation, provenance-linked retrieval and bilingual output can be combined into an advisory pipeline that runs in under 200 ms on a CPU, with no recommendation shown without a source and no prediction shown without a visual justification. Planned work is (i) an annotated in-situ field dataset from Swat, Mardan and Abbottabad for domain-adaptive fine-tuning, (ii) coverage of wheat stripe rust, sugarcane red rot and citrus canker, (iii) an open-set rejection class for non-target inputs, (iv) a Pashto speech interface, and (v) INT8 ONNX/TFLite export for native Android deployment.

Acknowledgment

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